cosmological constant, and if it's measurably different by any amount at all, we have a lot of Nobel Prizes. But even if you don't care about dark energy, or if you subscribe to the pessimistic view that we're fated to forever just narrow in on a garden-variety cosmological constant, dark energy surveys tend to be popular among astronomers of all stripes by doubling as all-purpose galaxy-gathering missions.

The upcoming LSST (Large Synoptic Survey Telescope), recently renamed the Vera C. Rubin Observatory (VRO), is a fantastic example. An 8.4-meter telescope on a high-desert mountain in Chile, VRO will take images of a few million supernovae and 10 BILLION galaxies, piecing together new images of the whole southern sky every few days. That kind of repeated coverage is great for supernova studies because it'll let us see the rise and fall of the brightness of each supernova over the several days during which the explosion is visible. But it's also great for studying galaxies, because it means you can stack up images night by night and see fainter and more distant galaxies than any other survey of its kind.

(As an aside, I recently attended a conference session about Planetary Defense in which the speakers were discussing the kinds of observations you need to spot potentially hazardous asteroids that might be on a collision course with our fragile little planet. The VRO will, at least for the southern sky, revolutionize our ability to pick these things up early, which might make it easier to find ways to stop them. I get a kick out of the idea that by attempting to understand the dark energy that will eventually destroy the universe, we might have a better chance of, on a much shorter timescale, saving the world.)

Whatever else its uses, the cosmological value of the VRO can't be overstated, if only because having massive piles of exquisite data gives us a very good chance of finding something new and surprising. According to Peiris, the VRO will be a game changer. "We are looking at the universe in a different way than what's been done before," she says. "Any time we've looked at the universe in a way that we haven't before, we learn new stuff."

VRO isn't the only new observational program to get excited about. There are a slew of other new telescopes and surveys coming up, each of which is poised to show us the cosmos in ways we've never seen before. Some of the most hotly anticipated are a class of new space telescopes like the James Webb Space Telescope (JWST), Euclid, and the Wide Field Infrared Survey Telescope (WFIRST), which will take deep images and